

PYTHON FULL STACK		
OBJECTIVE: The objective of this Python Full Stack Development course is to equip learners with comprehensive knowledge and practical skills required to design, develop, and deploy dynamic web applications. By mastering HTML, CSS, JavaScript, Python, Flask, and MySQL, students will be able to build end-to-end web solutions, handle front-end and back-end development, and manage databases efficiently. This course aims to provide a solid foundation in web technologies, foster problem-solving abilities, and promote best practices in web development.		
Duration:		
S.NO	CONTENT	HOURS (L/P)
Frontend Development (HTML, CSS, and JavaScript)		
	HTML	
1	Introduction to HTML What is HTML? HTML Text Editor	
2	HTML Basics HTML Tags HTML Attributes HTML Elements	
3	Text Formatting HTML Formatting HTML Heading HTML Paragraph	
4	Links and Images HTML Anchor (Links) HTML Image	
5	Tables and Lists HTML Table <u>HTML Lists</u> Ordered List Unordered List Description List	
6	Forms in HTML HTML Form Form Input Types Form Attributes	
7	Advanced HTML HTML with CSS HTML Classes and IDs HTML Iframes HTML JavaScript Integration HTML Comments	
8	HTML Layout and Responsive Design HTML Layout Basics HTML Responsive Techniques	
9	Miscellaneous HTML Color Names HTML Button onClick HTML Favicon HTML Login Form	
	CSS	
10	Introduction to CSS What is CSS? CSS Syntax CSS Selector Types	
11	Applying CSS Inline CSS Internal CSS External CSS CSS Comments	
12	CSS Properties Background Properties Border Properties B14Display Properties Font Properties Color and Text Properties Box Model (Margin, Padding, Width, Height) Positioning (Absolute, Relative, Fixed)	

13	Advanced CSS CSS Flexbox and Grid (justify-content, align-items) CSS Pseudo-classes CSS Transitions and Animations CSS Media Queries for Responsive Design	
14	Miscellaneous CSS Icons CSS Box-shadow and Text-shadow CSS List Styling CSS Navigation Bar Styling	
	JavaScript	
15	Introduction to JavaScript JavaScript Basics JavaScript Example	
16	Variables and Data Types JavaScript Variables JavaScript Data Types JavaScript Operators	
17	Control Flow Conditional Statements (if, switch) Loops (for, while)	
18	Functions and Objects JavaScript Functions JavaScript Objects JavaScript Arrays, Strings, Dates, Math, Numbers, Booleans	
19	DOM Manipulation Document Object Model (DOM) Accessing DOM Elements Manipulating DOM Elements InnerHTML vs InnerText	
20	Events and Validation JavaScript Events Overview Event Handling (addEventListener) Form Validation using JavaScript Email Validation Example	
21	Browser Object Model (BOM) Window Object History Object Navigator Object Screen Object	
22	Miscellaneous External JavaScript JavaScript Comments JavaScript Best Practices	
Backend Development (Python, Flask, and MySQL)		
	Python Programming	
23	Python Basics	
	i Python Features High-level language Interpreted language Object-Oriented Programming Dynamic typing Extensive standard library	
	ii Python History Introduction to Python Evolution of Python versions Major milestones	
	iii Python Applications Web Development Data Science Automation Artificial Intelligence and Machine Learning Game Development Desktop GUI Applications	
	iv Python Installation Downloading and installing Python Setting up environment variables Installing IDEs (PyCharm, VSCode, etc.)	
24	Python Fundamentals	
	i Python Example Writing and running your first Python program	

	ii	Python Variables Variable declaration Variable naming conventions Assigning values to variables	
	iii	Python Data Types Numbers (int, float, complex) Strings Lists Tuples Sets Dictionaries	
	iv	Python Keywords List of all keywords in Python Usage and examples	
	v	Python Literals String literals Numeric literals Boolean literals Special literals (None)	
	vi	Python Operators Arithmetic operators Comparison operators Logical operators Bitwise operators Assignment operators Identity operators Membership operators	
	vii	Python Comments Single-line comments Multi-line comments	
	viii	Control Flow Python If Else if statements if-else statements elif statements	
	ix	Python Loops while loop for loop pass break continue	
25		<u>Data Structures</u>	
	i	Python Strings String manipulation and methods	
	ii	Python Lists List operations and methods	
	iii	Python Tuples Tuple operations and methods	
	iv	Python List vs Tuple Differences between lists and tuples	
	v	Python Sets Set operations and methods	
	vi	Python Dictionary Dictionary operations and methods	
26		<u>Functions and Modules</u>	
	i	Python Functions Defining functions Function arguments and return values	
	ii	Python Built-in Functions Commonly used built-in functions	
	iii	Python Files I/O Reading and writing files	
	iv	Python Modules Importing and using modules Creating your own modules	
27		<u>Exception Handling</u>	
	i	Python Exceptions Handling errors with try-except Finally clause Raising exceptions	

28	Working with Dates	
i	Python Date Working with dates and times	
29	File Handling	
i	Write CSV File Writing data to CSV files	
ii	Read Excel File Reading data from Excel files	
iii	Write Excel File Writing data to Excel files	
30	Advanced Topics	
i	Python List Comprehension List comprehension syntax and examples	
ii	Python Collection Module Using namedtuple, deque, Counter, and defaultdict	
iii	Python Math Module Mathematical functions and constants	
iv	Python OS Module Interacting with the operating system	
v	Python Random Module Generating random numbers	
31	Object-Oriented Programming (OOP)	
i	Python OOPs Concepts Classes and objects Attributes and methods Encapsulation Inheritance Polymorphism Abstraction	
ii	Python Object Class Creating and using classes and objects	
iii	Python Constructors The init method Constructor overloading	
iv	Python Inheritance Single inheritance Multiple inheritance Multilevel inheritance	
v	Abstraction in Python Abstract classes and methods	
Python MySQL		
32	Environment Setup Setting up MySQL server Installing MySQL connector	
33	Database Connection Connecting to MySQL database using Python	
34	Creating New Database Creating and selecting a database	
35	Creating Tables Defining and creating tables	
36	Insert Operation Inserting data into tables	
37	Read Operation Querying and reading data from tables	
38	Update Operation Updating existing records	

Python Flask		
39	Introduction to Flask What is Flask? Setting up the Environment Installing Flask Your First Flask App Understanding the Project Structure	
40	Flask Basics Routing Creating routes for your application. URL Building Building dynamic URLs. HTTP Methods GET, POST, PUT, DELETE. Template Rendering with Jinja2 Rendering HTML templates. Static Files Serving CSS, JavaScript, and images.	
41	Templates and Forms Template Inheritance Using base templates. Jinja2 Logic Using conditions and loops in templates. Web Forms Creating and handling forms. WTForms Integration Simplifying form handling. Form Validation Validating form inputs.	
42	Working with MySQL Connecting to MySQL Setting up Flask with MySQL. CRUD Operations Creating, reading, updating, and deleting records.	
43	User Authentication User Registration Creating new user accounts. Login and Logout Managing user sessions. Password Hashing Securing passwords. Session Management Maintaining user sessions.	
44	Real time project	